Questions by topic: Energy & material cycles: natural vs human systems - Set 1 (55 marks)

Answer all questions in the spaces provided.

0 1 Table 1 compares nitrogen fixation by Rhizobium bacteria in legume root nodules with the Haber process.

Table 1

Feature	Biological fixation	Haber process
Energy source	Sunlight captured by the plant (carbohydrates)	Fossil fuels (natural gas / coal)
Temperature	Soil temperature (10-30 °C)	About 450 °C
Pressure	Atmospheric	200 atmospheres
Catalyst	Nitrogenase enzyme	Iron
Energy density of input	Low	High
Waste products	None - all products used	CO_2, NOx

0 1 . 1 Use Table 1 to explain how natural processes differ from human industrial processes in their use of energy.

[4 marks]

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0 1 . 2 Give **three** other natural processes driven by solar energy that human society depends on.

[3 marks]

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0 1 . 3 Explain how the principle of low-temperature natural processes is being applied in human systems.

[3 marks]

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0 2 . 1 Explain how natural material cycles differ from linear human systems of material use.

[4 marks]

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0 2 . 2 Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

[2 marks]

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0 2 . 3 Describe how the design of a product can move it from a linear to a cyclical system.

[4 marks]

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0 3 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 3 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 3 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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0 4 . 1 Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			
Use of materials			
Nature of wastes			
Temperature of processes			

0 4 . 2 Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

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0 4 . 3 Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

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0 4 . 4 Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

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0 5 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 5 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 5 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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0 6 . 1 Explain **three** features of natural systems that make them sustainable.

[3 marks]

0 6 . 2 Explain why the human use of mineral resources makes their sustainable exploitation much harder than the use of materials in a natural cycle.

[2 marks]

0 6 . 3 Explain **three** principles of the circular economy that should be applied when a new product is designed.

[3 marks]

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0 6 . 4 Explain why an activity that is sustainable for a small population may be unsustainable for a large one. Use **one** example.

[3 marks]

0 6 . 5 Explain why urban populations are less likely than rural communities to notice that a resource is being over-exploited, and give **two** consequences of this.

[4 marks]

15

END OF QUESTIONS